

Daily Science Anecdotes

Topic: What the Planes Didn't Show

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Imagine being handed data from hundreds of airplanes that had returned from dangerous missions.

Across the aircraft, the damage was not evenly distributed. Some areas showed many marks, while other areas showed very few.

The obvious conclusion seemed simple: protect the areas with the most damage.

But Abraham Wald, a statistician working with the Statistical Research Group at Columbia University during World War II, noticed something that was easy to miss.

The data contained only the airplanes that had survived.

And that changed everything.

The airplanes that returned were not a random sample of all the airplanes that had flown. They were the survivors. The airplanes that were lost were missing from the dataset completely.

So Wald asked a much more important question:

What can the missing airplanes tell us?

Suppose two parts of an airplane are compared. One part is frequently damaged on returning airplanes, while another part is rarely damaged.

At first, the first part looks more dangerous.

But think carefully.

If an airplane can be damaged in the first part and still return, that damage can appear in the data.

If damage to the second part is much more likely to prevent the airplane from returning, then those cases disappear from the dataset. We do not see them because the airplanes carrying them are no longer among the survivors.

In other words, **the absence of evidence in the surviving airplanes can itself be evidence of what happened to the missing ones.**

This is the heart of what later became a famous example of **survivorship bias**.

Wald's insight was not simply to "look where there were no holes." His actual work was more mathematical: he developed a method for estimating the vulnerability of different parts of an aircraft using information from surviving aircraft. His 1943 work was later reissued by the Center for Naval Analyses, and a 1984 paper by Marc Mangel and Francisco Samaniego examined Wald's methods in detail.

The deeper lesson is surprisingly useful outside aviation.

Imagine that you want to discover which students are most likely to succeed, so you study only students who graduated. Or imagine studying successful companies while ignoring the companies that disappeared. In both cases, the information you see may be systematically different from the information you would have seen if the missing cases were still present.

The problem is not necessarily that the data are wrong.

The problem is that **the data may have been filtered before you ever saw them.**

That is why survivorship bias is so dangerous: the numbers can be completely accurate and still lead to the wrong conclusion if we forget to ask who is missing.

Abraham Wald's story gives us a powerful rule for thinking about data:

Before asking what the data say, ask who the data left out.

Sometimes the most important information is not hidden inside the numbers we have.

It is hidden inside the numbers we never got to see.